



Longitudinal nationwide analysis of uveal melanoma in the United States 1995–2018

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Abstract

Purpose Uveal melanoma (UM) is the most common primary intraocular malignancy in adults, yet comprehensive nationwide epidemiologic data remain limited. This study aimed to provide an updated analysis of UM incidence, survival, and disparities in the United States (US) using near-complete population coverage.

Methods We analyzed data from the North American Association of Central Cancer Registries (NAACCR) and US Cancer Statistics (USCS) program between 1995 and 2018, covering 98–100% of the US population. Age-adjusted incidence rates (AAIRs) and relative survival were calculated using SEER*Stat and Joinpoint regression.

Results A total of 37,917 UM cases were identified, with an overall AAIR of 5.54 per million (95% CI: 5.48–5.59), showing a stable but slightly increasing trend (AAPC = 0.06). Incidence was highest in non-Hispanic whites (6.87 ppm), men (6.25 ppm), and those aged 75–79 years. Geographic variation was notable, with peak incidence in Oregon and Iowa. Most tumors (81.3%) were localized at diagnosis, and 27.4% of patients had another primary malignancy. 5- and 10-year relative survival rates were 83.3% and 73.9%, respectively, with worse outcomes in metastatic disease. A shift toward radiotherapy, especially brachytherapy, was observed, although survival gains remained limited.

Conclusions This nationwide study refines US UM incidence estimates and highlights persistent disparities by sex, race, age, and geography. The high prevalence of secondary malignancies underscores the need for genetic counseling and extended surveillance. Despite advances in imaging, local therapies (e.g., brachytherapy, proton therapy), and emerging systemic approaches including immunotherapy, survival improvements in advanced UM remain limited.

Keywords Uveal melanoma · Incidence · Survival · Ethnic disparities · Ocular oncology

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Abbreviations

AAIR	Age-adjusted incidence rate
US	United States
SEER	Surveillance and Epidemiology End Results Program
NAACCR	North American Association of Central Cancer Registries
NPCR	National Program of Cancer Registries
ICD-O-3	International Classification of Diseases-Oncology-3
CI	Confidence intervals
APC	Average percent-change

Introduction

Uveal melanoma is the primary ocular malignancy afflicting adults, with its incidence demonstrating considerable variability across global populations [1–5]. A nuanced

understanding of its epidemiological landscape is pivotal in shaping targeted preventative and therapeutic strategies. Historically, the data delineated by the Surveillance and Epidemiology End Results Program (SEER) have guided our comprehension of the disease's dynamics in the United States (US) [3, 6]. Nevertheless, the SEER database encompasses data from a mere 18 cancer registries, encapsulating 33% of the US populace, thus potentially precipitating under- or over-estimations in specific sub-populations and inducing fluctuations in the estimated incidence rates over time [7]. Disparities related to sex, race, socioeconomic status and residential setting posed significant challenges in prior research efforts [8].

To address these gaps, we leveraged data from the US Cancer Statistics Program (USCS) and the North American Association of Central Cancer Registries (NAACCR), representing nearly the entire US population. Our twofold objective was to refine nationwide uveal melanoma incidence estimates and examine net survival over two decades, excluding other mortality causes.

By drawing on extensive national databases, we anticipate more accurate incidence mapping, clearer insights into demographic and regional disparities, and a robust survival analysis. These findings will guide public health interventions, inform research priorities, and ultimately optimize patient outcomes in ocular oncology.

Methods

In this study, we adhered to the STROBE criteria for reporting cohort studies.

Data source and selection

This study embarked on a meticulous investigation into the incidences and trends of uveal melanoma, initially utilizing three principal databases to gather comprehensive data and annual variation between different databases (Accessed on 08.09.2021):

1. North American Association of Central Cancer Registries (NAACCR) Incidence and Survival Data-CiNA Analytic File (1995–2018) [9].
2. National Program of Cancer Registries (NPCR) and SEER Incidence—USCS Public Use Database (2001–2018), disseminated by the US Department of Health and Human Services, Centers for Disease Control and Prevention, and the National Cancer Institute in June 2021 [10].

3. SEER Research Data, 9 Registries, November 2020 Submission (1975–2018)—Integrated with County Attributes—Time-Dependent (1990–2018) Income/Rurality, spanning 1969–2019 Counties [7].

Subsequent analyses were orchestrated to calculate the incidence and survival, respectively, facilitating a coherent comparison with previously disseminated findings. Our selection criteria incorporated only registries aligned with the USCS quality benchmarks, encompassing a significant 98%–100% of the US demographic. Data from NPCR was used to outline the states-specific incidence rates.

Case identification and classification, inclusion and exclusion criteria

The ocular melanoma cases were identified using the International Classification of Diseases-Oncology–3 (ICD-O-3) [11]. We employed topography codes "69.0–69.9" and morphology code "7.1 melanoma", as delineated in the adolescence and young adults site recode/World Health Organization 2008 classification. Tumors labeled as C69.3 (choroid), and C69.4 (ciliary body and iris) were classified as 'uveal tumors'. Furthermore, tumors identified as C69.5 (retina, 0.5%) were grouped under choroidal tumors, acknowledging the previously reported misclassification [6]. Due to discrepancies across different SEER stage classifications, this study's regional extension encompasses direct orbital extension (outside the eye) and extension to regional lymph nodes. Patients with unknown age, survival status or cause of death were excluded from survival analysis.

Ethical considerations

Our study strictly adhered to the precepts outlined in the Good Clinical Practice Guidelines and the Declaration of Helsinki. Given that the US Cancer Registry publishes anonymized data, our study is classified as a nonhuman subject study, thereby waiving the necessity for Institutional Review Board (IRB) approval.

Data availability

Interested individuals can request the data from the NAACCR and NPCR/CDC/USCS. The processed data and query files are readily accessible on (<https://doi.org/10.5281/zenodo.5573791>). Moreover, interactive versions of the generated graphs can be retrieved from the authors' website: (https://bit.ly/UM_USA2021).

Software utilization

The data extraction and analysis were facilitated through the utilization of SEER*Stat Software (version 8.3.9), which was instrumental in calculating the age-adjusted incidence rates (AAIR) per million persons [12]. Subsequent data management was adeptly handled using Microsoft Excel [13]. Figure illustrations were crafted using Tableau Software (version 2023.02) [14] and OpenStreetMaps [15]. The survival trend analysis was executed using the JPSurv tool from NCI [16].

Statistical methods

Incidence rates were calculated per 1,000,000 individuals, aligning them to the 2000 US standard population (19 age groups—Census P25-1130) standards [17]. The set confidence intervals (CI) were 95%, with modifications influenced by the contributions of Tiwari et al. [18]. Consistent with NAACCR/CDC/NPCR regulations, we refrained from presenting counts and other estimates for groups harboring fewer than 16 patients to address reliability and privacy concerns.

We calculated confidence intervals, rate ratios, and ratio *p* values, facilitating a comprehensive comparison between study groups. Survival analysis was conducted for 28,312 uveal melanoma patients documented in the NAACCR database. Net survival, excluding other causes of death, was estimated using the relative survival actuarial method and the Ederer II method for cumulative expectation. The relative survival was deduced from the expected survival table "US 1970–2017 by individual year (White, Black, Other American Indian and Asian/Pacific Islander (AI/API), Ages 0–99, All races for Other Unspecified 1991 + and Unknown)". Trend analysis was conducted using the annual average percent-change (APC) endpoint and weighted least squares regression for two-year averages [19], considering *p* values less than 0.05 as significant.

Results

Patient demographics and incidence

We identified 37,917 patients with uveal melanoma in the NAACCR database (1995–2018), representing 2.7% of the 1,391,186 reported melanomas (Supplementary Table 1). Both NAACCR and NPCR–SEER sources revealed relatively stable uveal melanoma incidence over time, whereas the SEER9 dataset displayed more fluctuation (Supplementary Fig. 1A). Overall incidence was 5.54 per million (ppm; 95% CI: 5.48–5.59) for NAACCR and confirmed by NPCR–SEER data (5.54 ppm; 95% CI: 5.46–5.58), but with year-to-year variation in the age-adjusted incidence rate (AAIR; Fig. 1). Given the longer temporal coverage, NAACCR data were primarily used for descriptive analysis.

Choroidal melanoma was the predominant site, comprising 86.6% ($n=32,839$) of cases at 4.84 ppm (95% CI: 4.78–4.89). Iris/ciliary body melanomas accounted for 4768 cases (0.7 ppm; 95% CI: 0.68–0.72).

Demographic variations

Sex and ethnicity

Men exhibited a significantly higher AAIR (6.25; 95% CI: 6.16–6.34) than women (4.97; 95% CI: 4.90–5.05) ($p < 0.001$), consistently observed for both choroidal and ciliary body tumors. Non-Hispanic whites showed the highest incidence (6.87; 95% CI: 6.80–6.94), although Hispanic whites and African American populations had lower rates (Table 1).

Geographic distribution

State-specific analysis revealed that Oregon and Iowa had the highest incidence, whereas Hawaii, the District of Columbia, and North Dakota reported the lowest (Fig. 2). Rates also varied for ciliary body tumors. Urban–rural comparisons indicated incidences of 5.94 ppm (95% CI: 5.80–6.08) in nonmetropolitan areas and 5.46 ppm (95% CI: 5.40–5.52) in metropolitan regions (Table 1).

Age distribution

Peak incidence occurred among individuals aged 75–79 years (23.04 ppm) before declining in older cohorts (Supplementary Fig. 2). Ciliary body tumors showed a similar pattern in women aged 70–74 (2.95 ppm). Spindle cell melanomas presented at younger ages and at higher rates than epithelioid or mixed subtypes (data not shown).

Clinical characteristics

Staging and pathology

Most uveal melanomas were localized at diagnosis (81.3%, $n=30,817$). Regional extension was noted in 4.7% ($n=1765$), although 2.0% ($n=740$) demonstrated distant spread. Histopathologic confirmation was common (53.1%), though direct visualization and radiographic assessment were employed frequently (Supplementary Table 2). Approximately 80% of cases were categorized as malignant melanoma, NOS (ICD-O-3 code 8720/3). Of the microscopically confirmed tumors, 17.4% were spindle cell and 5.7% epithelioid, but excluding “melanoma, NOS” histology revealed spindle cell types in 46.7% of tumors (Supplementary Table 3).

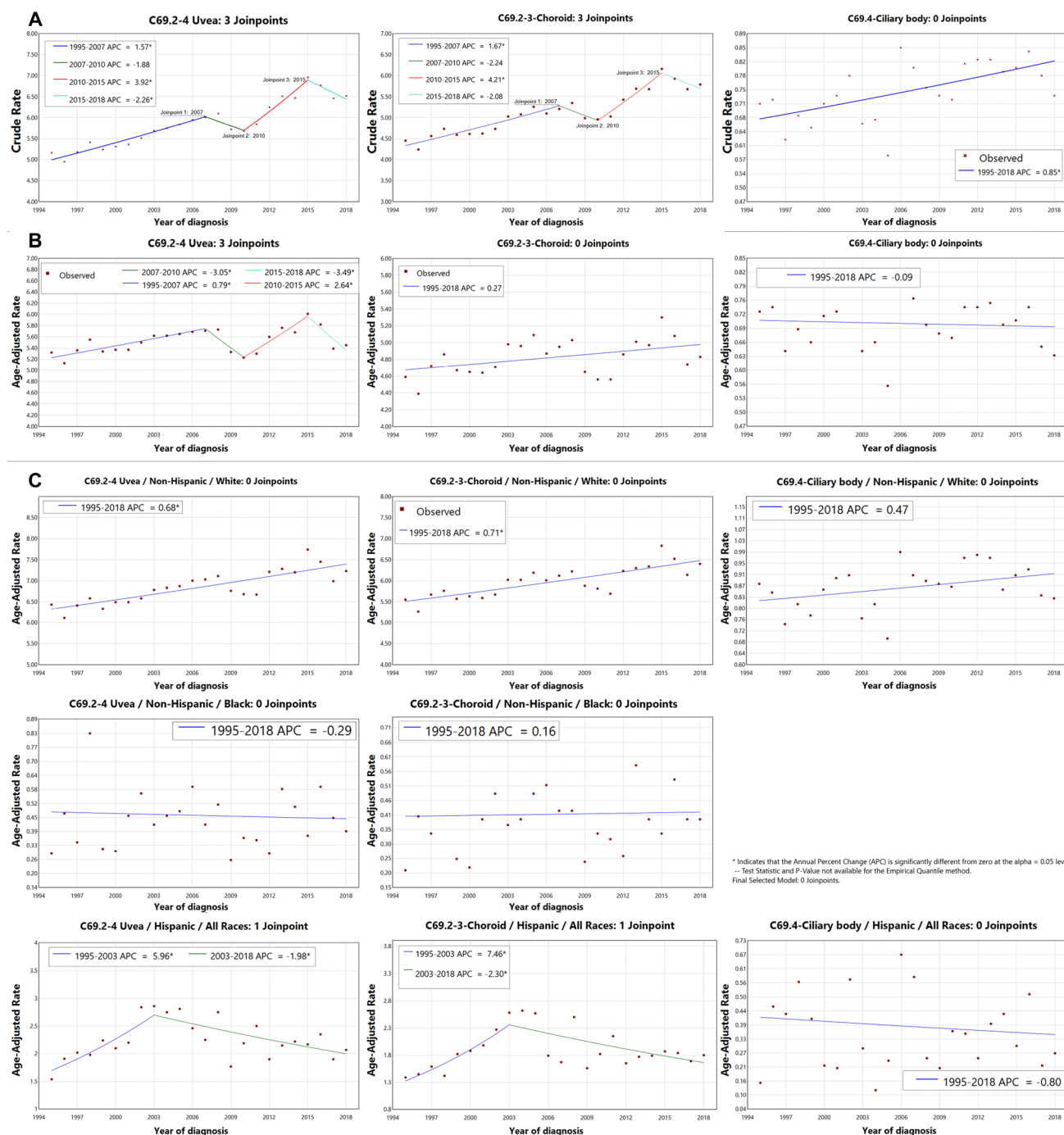


Fig. 1 Incidence rates of Uveal melanoma. **A.** Crude Rates per site B&C. **B.** Age-Adjusted, per site (B) and per site, race and origin (C)

Second malignancies

A notable 27.4% ($n=10,403$) of patients had additional primary malignancies, with 12.3% ($n=4,669$) experiencing a prior cancer and 15.1% ($n=5,734$) developing a subsequent malignancy (Supplementary Table 2). Key demographic and clinical features did not differ markedly between single- and multiple-cancer cohorts (data not shown).

Relative survival

Overall and anatomical sites

Five-year relative survival for uveal melanoma was 83.29% (95% CI: 82.67–83.88), and ten-year survival was 73.91% (95% CI: 73.04–74.76). Patients with localized disease fared significantly better (5-year: 85.85%, 10-year:

Table 1 Age-adjusted Incidence rates of uveal melanoma according to laterality, sex, race, region, and urban living status.

Laterality	C692—3: Choroid			C694: Ciliary body			Uvea		
	Rate (95%CI)	N	%	Rate (95%CI)	N	%	Rate (95%CI)	N	%
Total	4.84 (4.78–4.89)	33,149		0.7 (0.68–0.72)	4768		5.54 (5.48–5.59)	37,917	
Unilateral	4.79 (4.74–4.85)	32,869	99.2%	0.69 (0.67–0.71)	4705	98.7%	5.49 (5.43–5.54)	37,574	99.1%
Bilateral	*	6	0.0%	*	2	0.0%	*	8	0.0%
Unknown	0.04 (0.04–0.04)	274	0.8%	0.01 (0.01–0.01)	61	1.3%	0.05 (0.04–0.05)	335	0.9%
Sex									
Male	5.49 (5.41–5.57)	17,296	52.2%	0.76 (0.73–0.79)	2370	49.7%	6.25 (6.16–6.34)	19,666	51.9%
Female	4.31 (4.24–4.38)	15,853	47.8%	0.66 (0.63–0.69)	2398	50.3%	4.97 (4.9–5.05)	18,251	48.1%
NHIAv2 Race									
Hispanic—All—	1.89 (1.78–2)	1192	3.6%	0.34 (0.3–0.4)	225	4.7%	2.23 (2.11–2.36)	1417	3.7%
Races									
Hispanic—White	1.8 (1.69–1.91)	1128	3.4%	0.33 (0.28–0.38)	215	4.5%	2.13 (2.01–2.25)	1343	3.5%
Hispanic—Black	0.03 (0.02–0.05)	22	0.1%	0.01 (0–0.02)	6	0.1%	0.04 (0.03–0.06)	28	0.1%
Hispanic—Unknown		42	0.2%		4	0.1%		46	0.1%
Non-Hispanic—White	6 (5.93–6.07)	30,960	93.4%	0.87 (0.84–0.89)	4410	92.5%	6.87 (6.8–6.94)	35,370	93.3%
Non-Hispanic—Black	0.4 (0.35–0.45)	281	0.8%	0.05 (0.03–0.07)	32	0.7%	0.44 (0.39–0.5)	313	0.8%
Non-Hispanic—Other	0.74 (0.65–0.84)	255	0.8%	0.12 (0.09–0.17)	43	0.9%	0.86 (0.76–0.97)	298	0.8%
Non-Hispanic—Unknown	*	442	1.3%	*	57	1.2%	*	499	1.3%
Unknown—NHIA	*	19	0.1%	*	1	0.0%	*	20	0.1%
Metro/Non-Metro (2003 Beale)									
Metropolitan	4.77 (4.71–4.83)	26,957	81.3%	0.69 (0.67–0.71)	3874	81.3%	5.46 (5.4–5.52)	30,831	81.3%
Non-Metropolitan	5.18 (5.05–5.31)	6170	18.6%	0.76 (0.71–0.81)	891	18.7%	5.94 (5.8–6.08)	7061	18.6%
Unknown	3.83 (2.39–5.8)	22	0.1%	0.56 (0.11–1.59)	3	0.1%	4.39 (2.83–6.47)	25	0.1%

77.21%) than those with distant metastases (5-year: 25.48%, 10-year: 15%). Choroidal melanoma had generally better outcomes than ciliary body disease, and iris melanomas ($n=418$) exhibited the highest five-year survival (100%). Detailed survival by stage showed a stepwise decline from localized (5-year: 89.59%, 10-year: 77.23%), to regional (5-year: 65.02%, 10-year: 51.85%), and distant disease (5-year: 24.83%, 10-year: 14.63%) (Fig. 3).

Sex and ethnicity disparities

Women had slightly superior five-year relative survival (84.47%; 95% CI: 83.61–85.29) compared to men (82.17%; 95% CI: 81.29–83.02). Patients of Black race had marginally lower survival (82.18%; 95% CI: 75.02–87.46) than their white counterparts (83.05%; 95% CI: 82.42–83.66). American Indian/Alaska Native and Asian/Pacific Islander groups had five-year rates of 88.75% (68.28–98.51) and 80.05% (71.70–86.16), respectively.

Trends

Incidence

Unlike conjunctival melanomas (annual rate reduction of 1.4%; $p=0.013$), uveal melanoma incidence showed no significant annual decline but rather a modest upward trend (AAPC = 0.06) (Fig. 1). Meanwhile, nonocular melanomas increased by 1.74% annually ($p<0.001$). An average 5.1% yearly reduction was noted for “eye–not otherwise specified” tumors ($p<0.001$, data not shown).

Diagnosis

Diagnostic approaches shifted substantially, with radiographic confirmation rising by 4.14% annually for choroidal melanomas ($p<0.001$) and histologic confirmations declining (AAPC = -0.59% for ciliary body, $p<0.001$) (Supplementary Fig. 3, Supplementary Table 4). Biomarker-based diagnostics experienced a mild increase late in the study period.

Treatment

For choroidal tumors, a significant rise in nonsurgical management was documented (AAPC = 4.2%, $p<0.005$), whereas enucleation decreased slightly (AAPC = -0.39% , $p=0.42$) (Supplementary Fig. 4, Supplementary Table 5). Brachytherapy and beam irradiation both increased significantly (AAPC = 4.14% and 1.73%, respectively; $p<0.005$),

although the proportion receiving no radiation fell by 1.46% annually ($p<0.005$) (Supplementary Table 6).

Survival

Five-year relative survival for uveal melanoma remained stable at approximately 83% (95% CI: 82.67–83.88) across the study period, with no statistically significant temporal trend (Annual Percent Change [APC]: 0.0%). Joinpoint regression modeling did not identify any significant inflection points (Joinpoint = 0, SE: 0.00%). Bayesian Information Criterion (BIC = 171.02) and Akaike Information Criterion (AIC = 113.32) further supported the model without additional joinpoints. Similarly, no significant trends were detected in 10-year survival (Fig 4).

Discussion

Uveal melanoma is the most prevalent primary intraocular malignancy in adults, a trend substantiated by our comprehensive study spanning 1995 and 2018. This malignancy accounted for 2.63% of all melanomas and 87% of ocular melanomas, underscoring its significant representation in the spectrum of melanomas [20]. The present study embarks on an ambitious journey to elucidate the nuanced trends in the incidence rates of uveal melanoma within the US populace. By offering a panoramic view grounded in substantial patient data, this study aims to delineate the nationwide incidence rates with an unmatched level of precision and coverage, thereby filling a notable gap in the existing literature and providing a benchmark for future research endeavors in the field of ocular oncology.

In the landscape of uveal melanoma research, several studies have ventured to dissect the incidence rates on a global and US scale. However, limited population scopes have often constrained these investigations, potentially infusing bias into their findings [1, 2, 21–25]. Contrasting this, our study pioneers in leveraging a dataset that mirrors the entire US demographic, significantly mitigating the biases observed in previous studies that stemmed from the population structure and reporting delays in SEER-covered regions [26–29].

A meticulous comparative analysis with seminal works by Andreoli et al. and Aronow et al., which scrutinized 7043 and 4999 cases, respectively, echoes the gradual elevation in the mean age at diagnosis over the years. This trend is postulated to be a consequence of enhanced life expectancy coupled with the increasing frequency of ophthalmic examinations among the elderly demographic. Our study further nuances this observation by highlighting a conspicuous peak in the AAIR within the 75–79 age bracket, succeeded by a subsequent decline [6, 30, 31].

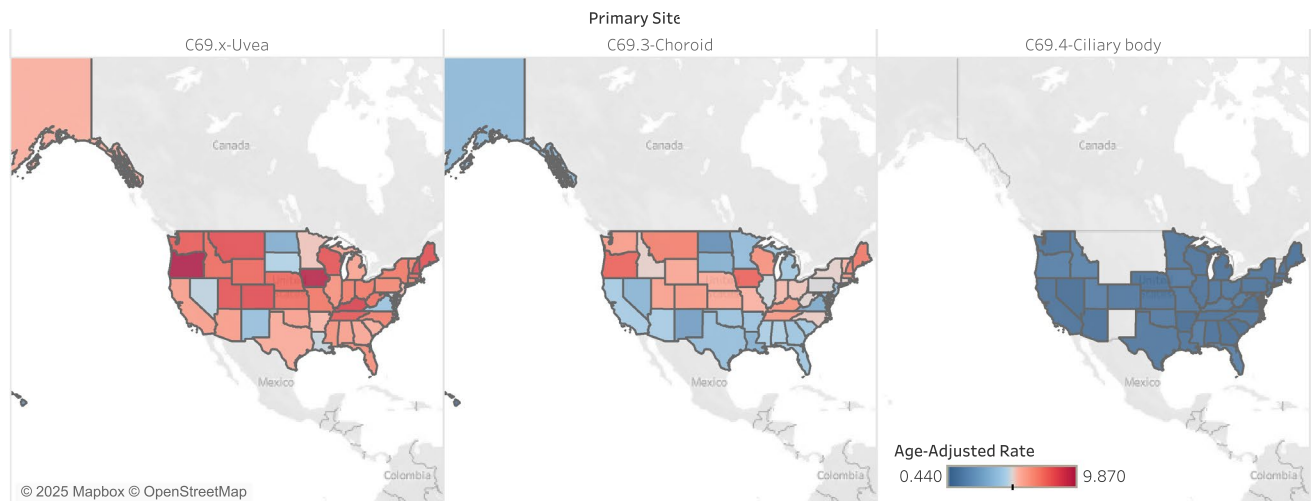


Fig. 2 Age-adjusted incidence rates in the states per site (per million), standardized to the 2000 U.S. standard population (from US Cancer Statistics Program 2001–2018)

Moreover, our endeavor distinguishes itself by being the first to highlight a decrement in incidence rates among the Spanish, Hispanic, and Latino demographics, offering fresh perspectives and avenues for further investigation. This substantial dataset, encompassing the period between 2001 and 2018, facilitates a robust and accurate trend analysis, standing in stark contrast to preceding studies that were limited by smaller population fractions and shorter durations, thus presenting a more precise reflection of the epidemiological landscape of uveal melanoma [5, 32].

Our study has been discerned that men manifest a higher incidence rate as compared to women, with non-Hispanic whites showcasing a markedly higher incidence rate than Hispanic whites and African Americans, corroborating the findings of earlier studies that reported similar disparities without clear statistical significance [33, 34]. Furthermore, the survival analysis divulged a promising five-year relative survival rate of 83.15% and a ten-year survival rate of 73.84%, offering pivotal prognostic information that underscores the necessity for long-term follow-up and monitoring of uveal melanoma patients [35–37].

Despite a paradigm shift from enucleation to globe-preserving therapies and refinements in local tumor control using proton therapy, we observed no significant improvement in long-term survival over the study period. This aligns with prior reports indicating that survival outcomes are predominantly determined by metastatic potential rather than local tumor management. The stable survival trends (five-year APC: 0.0%) suggest that systemic disease control remains a critical unmet need. Although recent therapeutic advances, including immunotherapy, hold promise for improving metastatic uveal melanoma outcomes, these approaches were not widely adopted within our study

window and thus their impact was not reflected in our data. Future population-based analyses incorporating newer systemic treatment eras will be essential to evaluate their true effect on survival.

In addition to demographic nuances, the study unveils notable geographic disparities in the incidence rates of uveal melanoma across the United States. A meticulous analysis indicated higher incidence rates in the Midwest and Northeast regions, contrasted by a lower rate observed in the South. These disparities, potentially influenced by a confluence of factors, including varied exposure to risk factors, accessibility to healthcare, and genetic predispositions, warrant further exploration to develop targeted interventions [38, 39]. Furthermore, state-specific analysis illuminated stark variations in incidence rates, with certain states manifesting markedly higher rates, underscoring the necessity for a deeper investigation into the underlying causative factors.

Regarding clinical attributes, most uveal melanoma cases were identified at the 'local' stage, succeeded by 'regional extension' and 'distant extension' cases. Microscopic confirmation emerged as the predominant diagnostic modality, followed by direct visualization and radiography, accentuating the significance of early detection and precise diagnosis in enhancing patient outcomes. These findings pave the way for a renewed focus on the evolution of diagnostic methodologies and their implications on the management and prognosis of uveal melanoma [40–42]. Moreover, our findings underscore the critical importance of ensuring data completeness and quality in cancer registry documentation, particularly in light of the high rate of missing data we observed.

The temporal analysis embedded in the study points towards a nonsignificant annual percentage change in the incidence rates

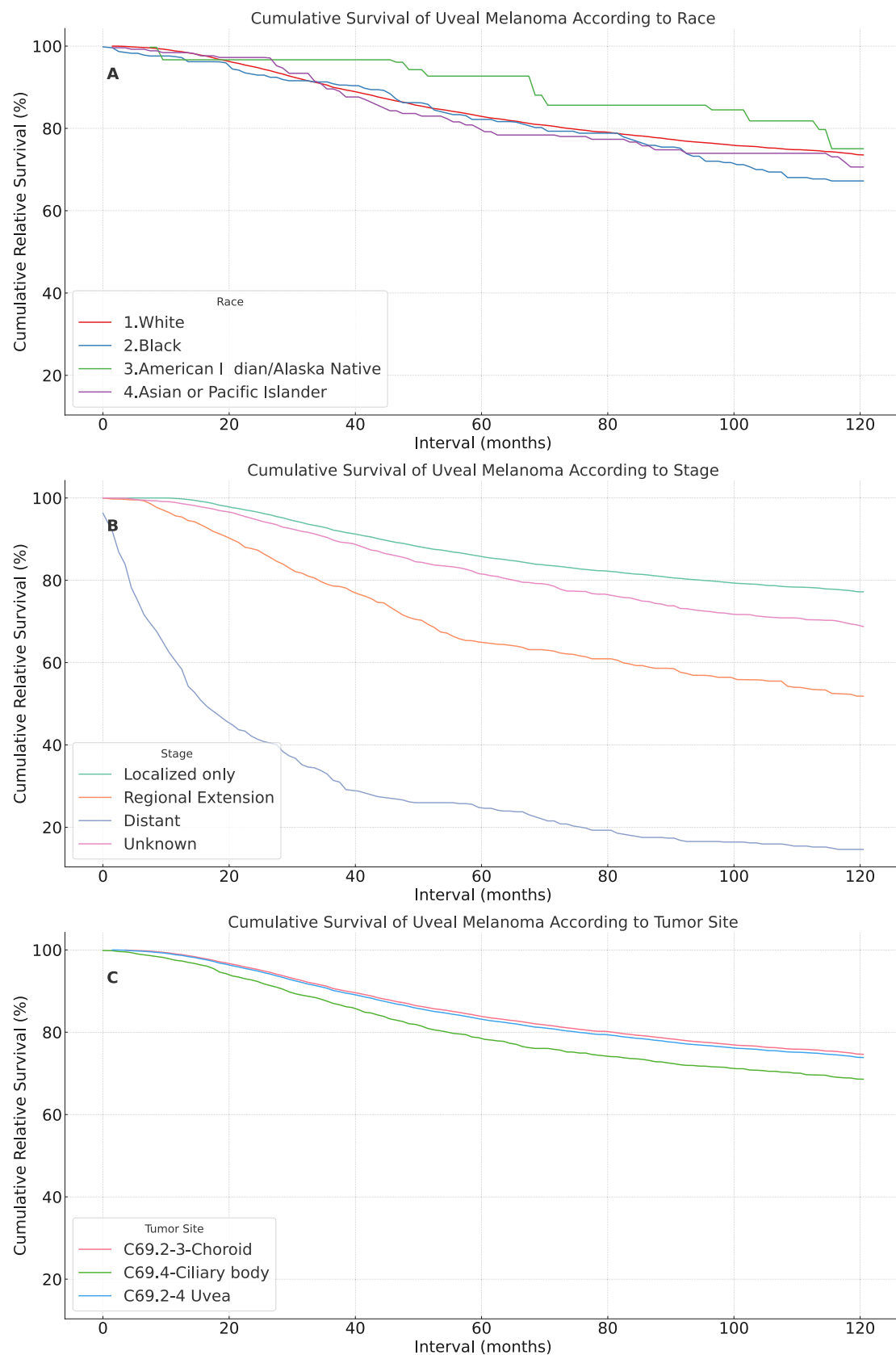


Fig. 3 Cumulative relative survival per race (A), stage (B) and site (C)

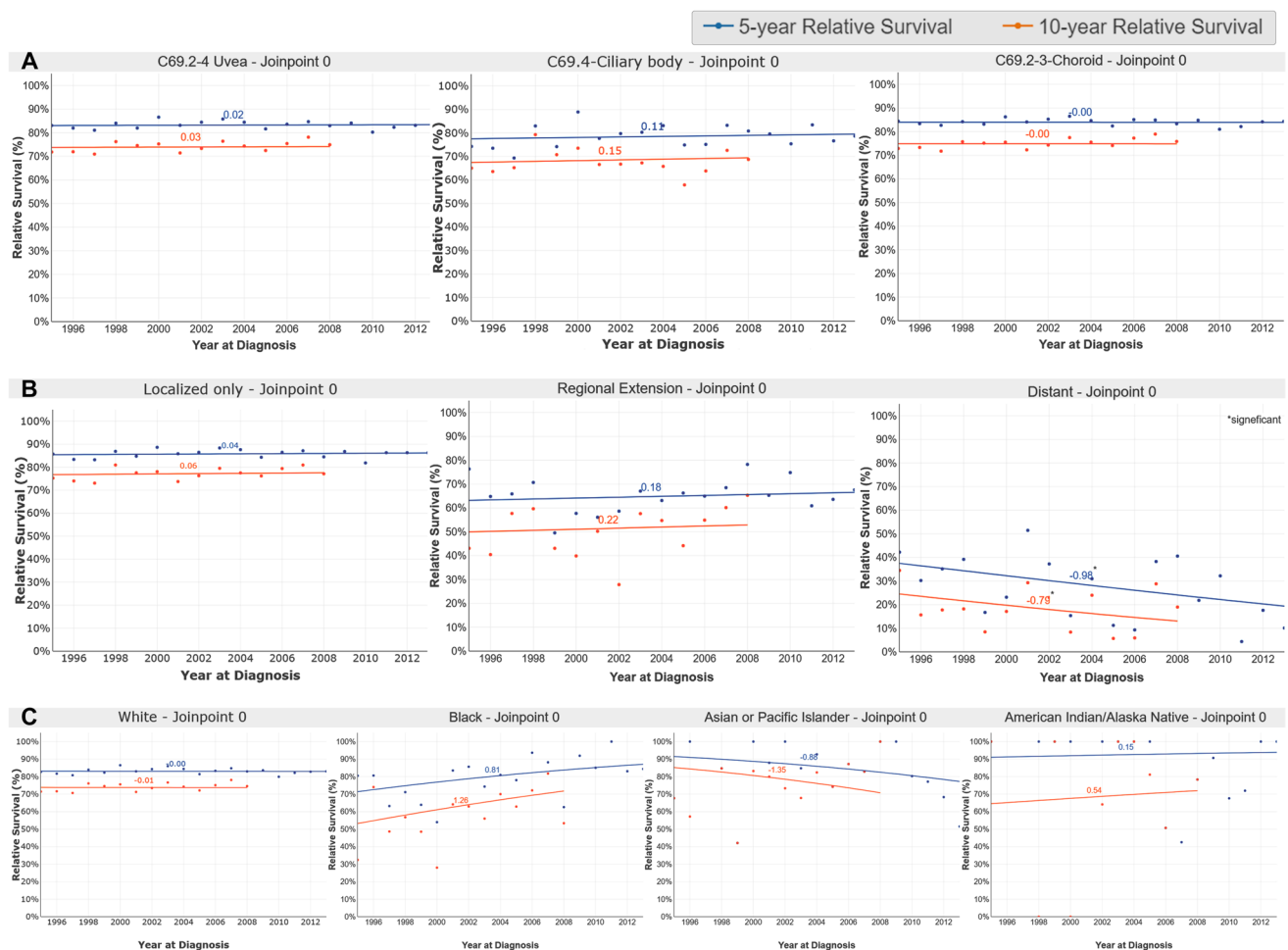


Fig. 4 The trends of relative survival using JoinPoint Survival Analysis according to **A** Site, **B** Stage, **C** Race. None of the subgraphs showed significant results

for uveal melanoma over the investigated period. This facet suggests relative stability in the incidence of uveal melanoma over time, albeit accompanied by significant variations in diagnostic modalities, tumor staging and treatment trends, thereby indicating a dynamic shift in clinical practices and disease management over the years. However, these active changes did not reflect on the therapy outcome.

A significant revelation from our study is the Changing landscape of diagnostic trends, which have witnessed a shift towards an increasing reliance on clinical assessments as opposed to histological confirmation. This transition is particularly noticeable in diagnosing ciliary body tumors, where a decline in pathological confirmation has been observed in favor of clinical diagnosis. Furthermore, we observed that approximately 80% of tumors were classified as malignant melanoma, NOS, highlighting the potential loss of histopathologic resolution in recent decades. This finding underscores the critical need for standardized histologic

and molecular subclassification in future registry practices. Although advancements in diagnostic techniques, such as ultrasonography and optical coherence tomography, have facilitated early detection and improved treatment outcomes, this trend also raises concerns. Specifically, the increasing incidence of regional extension of malignancies noted for the first time may result from this shift in diagnostic preference, possibly due to the inability to detect regional extensions earlier [40–43]. On the other hand, it may represent an improvement of the diagnostic methods that were unable previously to detect disease extension until distant extension occurs. We acknowledge that the inclusion of C69.5-coded tumors as choroidal melanoma, although supported by prior evidence, may introduce minor classification bias; however, this strategy was chosen to more accurately capture true uveal melanoma burden as per previous reports [6, 31].

Our findings also highlight the complexity of managing patients with uveal melanoma, particularly in relation to the

occurrence of secondary malignancies. Approximately 27% of patients were diagnosed with additional primary malignancies, underscoring the importance of comprehensive surveillance and integrated care strategies. Notably, comparative analyses between uveal melanoma-only cohorts and those with multiple primary cancers have been conducted and published by our group as well as by others [44, 45]. In that study, we demonstrated significant differences in age, sex, tumor characteristics, and survival between these groups, supporting the rationale for extended follow-up and genetic counseling. Specifically, UM patients with additional malignancies tended to be older, had a higher proportion of males, and showed distinct survival advantages in certain subgroups. Given that analysis was beyond the present study's scope, we refer readers to that comprehensive publication for further details. Our current findings nonetheless reinforce the need for heightened vigilance and potential genetic assessment, as studies indicate that up to 9% of unselected UM patients harbor germline mutations in high- and low-penetrant cancer genes. The National Comprehensive Cancer Network (NCCN) guidelines recommend referral for genetic evaluation in patients at high risk for cancer predisposition syndromes.

The data also signal a potential need to reassess and possibly revise existing treatment protocols and follow-up strategies. The observed increase in the incidence of regional extension of the disease, which could be attributed to improvements in diagnostic tools, suggests a need for a stricter definition of the population at risk and a revision of follow-up protocols. In conjunction with the relatively high rates of previously reported distant metastasis, this aspect necessitates further studies to unravel other causes behind this increase and to explore any associations with recent changes in clinical practice [6, 31, 40, 46].

Despite its breadth and nationwide representativeness, our study is constrained by several inherent limitations related to cancer registry data. First, the use of high-level histologic codes (e.g., malignant melanoma NOS, ICD-O-3 code 8720/3, representing ~80% of cases) limits the ability to accurately assess the distribution and prognostic implications of detailed subtypes such as spindle, epithelioid, and mixed cell types [32]. This classification ambiguity introduces potential misclassification bias and underscores the necessity of integrating standardized histopathologic and emerging molecular subclassification into future registry protocols. Second, staging variables may suffer from inconsistencies and underreporting across different registries, particularly regarding regional and distant extension, which may influence survival analyses and patterns of care. Third, treatment data in registries capture broad categories (e.g., surgery, radiation) without distinguishing specific approaches such as plaque brachytherapy versus proton beam or stereotactic radiosurgery. This lack

of granularity impedes a detailed evaluation of therapeutic advances and their contribution to survival outcomes. Future studies leveraging prospective multicenter cohorts or augmented registry data with detailed procedural and molecular annotations will be essential to refine our understanding of treatment impact and prognostic stratification in uveal melanoma.

Further global comparisons of incidence and survival are warranted, alongside efforts to identify risk factors and develop prevention strategies. Improved diagnostic precision within existing data repositories will allow more refined analyses, contributing to targeted public health interventions to address disparities.

In summary, this work is a seminal contribution to ocular oncology, presenting a nationwide analysis of uveal melanoma incidence in the US. Although constrained by certain limitations, it lays a foundation for ongoing research and management strategies by highlighting key demographic and racial differences. Beyond serving as a benchmark for future large-scale studies, it provides a framework for policymakers to optimize resource distribution in ocular oncology. Through continued interdisciplinary collaboration and focused research, these findings are expected to drive advances that enhance both clinical outcomes and patient-centered care.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10147-025-02870-7>.

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Author contributions Ahmad Samir Alfaar was responsible for the Conceptualization, Methodology, Formal Analysis, Investigation, Writing of the Original Draft, and Visualization. Mohamed H Abdel-Rahman and Moataz Hamed Osman both contributed to the investigation and were involved in Writing—Review & Editing. All authors have revised and accepted the final manuscript.

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Data availability No preprints were deposited elsewhere. Supporting materials can be found on <https://doi.org/10.5281/zenodo.5573791> and https://bit.ly/UM_USA2021

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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